

Feedback — How Did Yeast Become a Wine Maker?

(Quiz 1)

[Help Center](#)

Thank you. Your submission for this quiz was received.

You submitted this quiz on **Sat 16 May 2015 1:31 AM PDT**. You got a score of **10.00** out of **10.00**.

Question 1

True or False: Most genes in DeRisi's 6400-gene dataset are likely involved in the diauxic shift.

Your Answer	Score	Explanation
☐ True		
☒ False	✓ 1.00	
Total	1.00 / 1.00	

Question 2

True or False: In practice, the number of clusters of biological data is typically not known in advance.

Your Answer	Score	Explanation
☒ True	✓ 1.00	
☐ False		
Total	1.00 / 1.00	

Question 3

True or False: Center placement in the k -Centers Clustering Problem is generally more sensitive to outliers than center placement in the k -Means Clustering Problem.

Your Answer		Score	Explanation
☒ True	✓	1.00	
☐ False			
Total		1.00 / 1.00	

Question 4

How many ways are there to cluster 4 points into two clusters?

You entered:

7

Your Answer		Score	Explanation
7	✓	2.00	
Total		2.00 / 2.00	

Question 5

True or False: The Lloyd algorithm is not guaranteed to converge for every collection of data points and every choice of initial centers.

Your Answer		Score	Explanation
☐ True			
☒ False	✓	1.00	
Total		1.00 / 1.00	

Question 6

Compute $MaxDistance(Data, Centers)$ for the following *Data* and *Centers*:

Data: (2, 6), (4, 9), (5, 7), (6, 5), (8, 3)

Centers: (4, 5), (7, 4)

You entered:

Your Answer		Score	Explanation
4	✓	1.00	
Total		1.00 / 1.00	

Question 7

Compute $Distortion(Data, Centers)$ for the following *Data* and *Centers*:

Data: (2, 6), (4, 9), (5, 7), (6, 5), (8, 3)

Centers: (4, 5), (7, 4)

You entered:

Your Answer		Score	Explanation
6	✓	1.00	
Total		1.00 / 1.00	

Question 8

Give the center of gravity of the following (three-dimensional) data points. Enter your answer in the form (x, y, z).

(17, 0, -4), (3, 14, 23), (9, 7, 16), (7, 3, 5)

You entered:

(9, 6, 10)

Your Answer		Score	Explanation
(9, 6, 10)	✓	2.00	
Total		2.00 / 2.00	